Shaik Abdul Sami

Software Engineer | IIIT Dharwad 2027
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Professional Summary

Final-year B.Tech CSE student at IIIT Dharwad who builds and ships end-to-end software: REST APIs, full-stack web applications, and ML-integrated systems. Built and deployed three projects with automated tests, CI/CD, and JWT-secured APIs. Strong in Python and JavaScript, data structures and algorithms, and Git-based workflows.

Education

Indian Institute of Information Technology Dharwad, Karnataka **2023 – 2027**
B.Tech, Computer Science and Engineering Final Year, No Active Backlogs

Technical Skills

Languages: Python, JavaScript, C++, SQL, HTML5, CSS3
Backend: Node.js, Express.js, FastAPI, Flask, REST API Design, JWT Authentication
Frontend: React.js, Tailwind CSS, Responsive Design, Leaflet.js
Databases: MongoDB, MongoDB Atlas, MySQL
Testing and DevOps: Pytest, Postman, Git, GitHub Actions (CI/CD), Docker, Vercel, Railway, Linux
AI and ML: PyTorch, Sentence-Transformers, Embeddings and Semantic Search, Contrastive Learning, RAG (Retrieval-Augmented Generation), LLM API Integration (Anthropic API), Scikit-learn, Pandas, NumPy, GeoPandas
Core CS: Data Structures and Algorithms, Object-Oriented Programming, SDLC, Agile

Projects

GeoSafe AI – Smart Land Analyzer [GitHub] *Python, Scikit-learn, GeoPandas, Leaflet.js, Docker*

- Built a full-stack geospatial app that classifies land safety (Low / Medium / High) using a Random Forest model with OpenStreetMap, Natural Earth, and NASA SRTM data; approx. 90% accuracy.
- Designed a Python REST API with an R-tree spatial index and in-memory cache, returning results in under 500 ms per request.
- Added rule-based overrides for roads, restricted zones, and land-use conflicts; visualised results on an interactive Leaflet.js map; containerised with Docker Compose.

My-Jewellery – E-Commerce Platform [GitHub] *React.js, FastAPI, MongoDB Atlas, Pytest*

- Built and deployed a jewellery catalogue and enquiry platform: React.js on Vercel, FastAPI on Railway, MongoDB Atlas database.
- Implemented a real-time pricing engine (metal value, making charges, GST) driven by gold and silver rates managed from an admin dashboard.
- Secured the API with JWT, bcrypt, SlowAPI rate limiting, and Pydantic validation; wrote 12 Pytest tests with a mocked database, run by a GitHub Actions CI/CD pipeline.

Smart-Folio – Full Stack Portfolio CMS [GitHub] *React.js, Node.js, Express.js, MongoDB*

- Built a portfolio CMS with a React.js frontend, Node.js/Express.js REST API, and MongoDB (Mongoose) database.
- Implemented two-token JWT authentication (short-lived access token plus HttpOnly refresh cookie), bcrypt hashing, rate limiting (5 login attempts per 15 minutes), Helmet.js headers, and express-validator input validation.
- Built an admin dashboard with CRUD, drag-and-drop reordering (dnd-kit), and visibility toggles so content updates go live without code changes.

Taste Embedding Learning – Recipe Similarity Search [GitHub] *PyTorch, Sentence-Transformers, FastAPI, RAG, Anthropic API*

- Co-developed a recipe similarity engine: trained a projection head with a contrastive loss over `all-MiniLM-L6-v2` embeddings of 10,000 recipes, raising category match@10 from 0.599 to 0.676 on 2,000 held-out recipes (random baseline 0.181).
- Blended raw and learned embeddings after finding the learned model ranked dosa closer to a burger than pizza; the blend kept the intended ordering at 0.679 match@10.
- Built a FastAPI service and chat UI implementing RAG: Claude (Anthropic API) classifies user intent, the embedding index retrieves matching recipes, and Claude generates the response from the retrieved results.